Task 5: Capture and Analyze Network Traffic Using Wireshark

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Task: Task 5 – Network Traffic Analysis

Tool Used: Wireshark 4.6.9

Operating System: Windows

1. Objective

To capture live network traffic, identify common network protocols, and analyze packet details using Wireshark.

2. Tools Used

- Wireshark 4.6.9
- Windows operating system
- Wi-Fi network connection

3. Procedure

1. Installed Wireshark on Windows.
2. Selected the active Wi-Fi network interface.
3. Started capturing live network packets.
4. Generated traffic by using the internet.
5. Stopped the capture after collecting packets.
6. Applied display filters to analyze DNS, TCP, and TLS traffic.

7. Saved screenshots of the packet analysis.

8. Saved the capture file as "task5_network_capture.pcapng".

4. Protocols Identified

DNS (Domain Name System): Used to resolve domain names into IP addresses. DNS packets were observed using the "dns" display filter.

TCP (Transmission Control Protocol): Provides reliable communication between network devices. TCP packets were observed using the "tcp" display filter.

TLS (Transport Layer Security): Helps protect data exchanged over encrypted connections. TLS packets were observed using the "tls" display filter.

ICMPv6: Used for IPv6 network discovery and control messages. ICMPv6 packets were also visible in the capture.

5. Results

A total of 650 packets were displayed in the saved capture. DNS, TCP, TLS, and ICMPv6 traffic were identified. Display filters helped isolate specific protocol traffic for analysis.

6. Conclusion

This task provided practical experience in capturing live network traffic, applying Wireshark display filters, identifying network protocols, and saving packet captures for further analysis.

7. Deliverables

- "task5_network_capture.pcapng"
- "1-live_capture.png"
- "2-dns_filter.png"
- "3-tcp_filter.png"
- "4-tls_filter.png"
- "Task5_Network_Traffic_Analysis_Report.docx"